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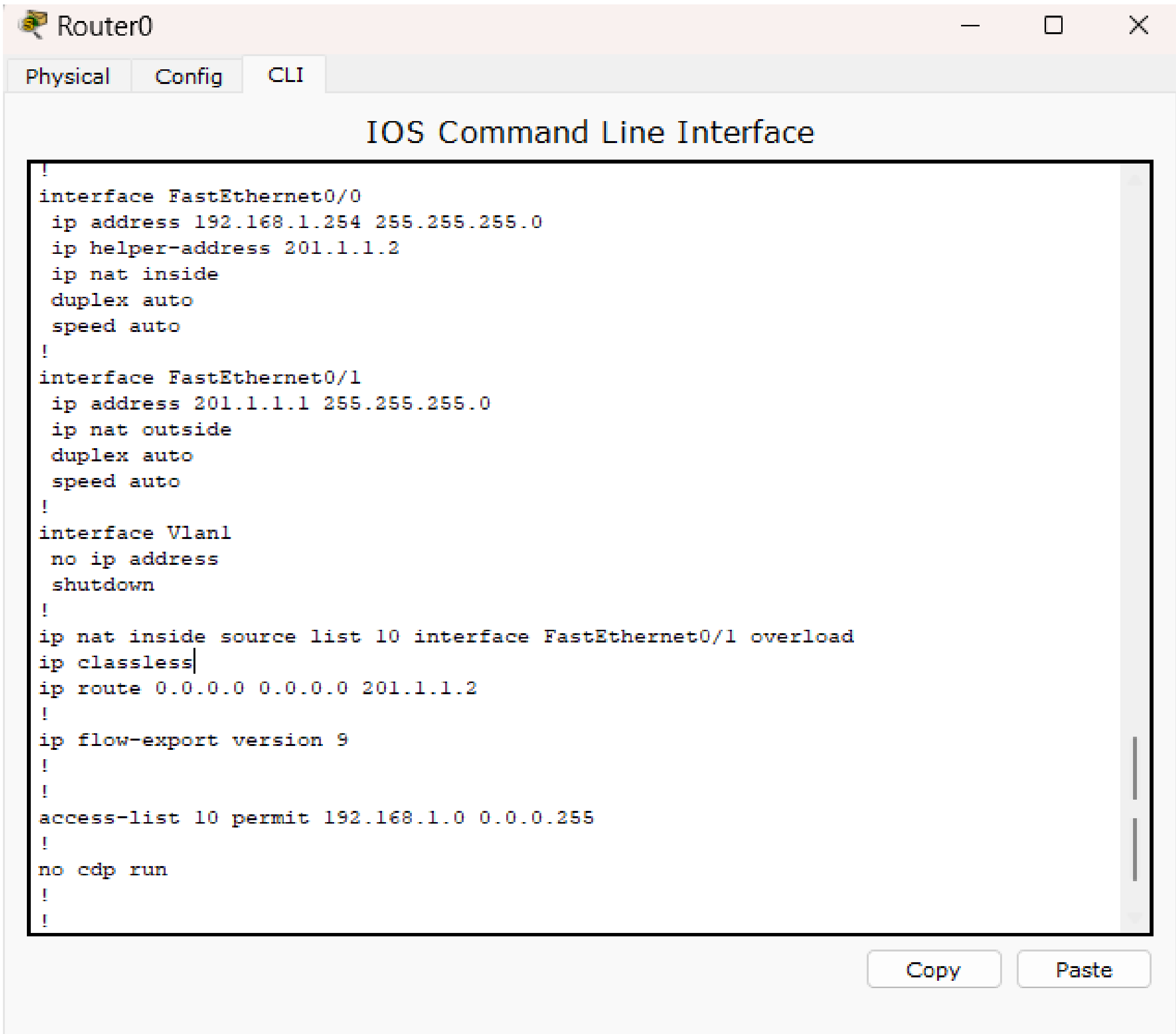
Question 1

Question A




```
!
interface FastEthernet0/0
 ip address 192.168.1.254 255.255.255.0
 ip helper-address 201.1.1.2
 ip nat inside
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 201.1.1.1 255.255.255.0
 ip nat outside
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
ip nat inside source list 10 interface FastEthernet0/1 overload
ip classless
ip route 0.0.0.0 0.0.0.0 201.1.1.2
!
ip flow-export version 9
!
!
access-list 10 permit 192.168.1.0 0.0.0.255
!
no cdp run
!
!
```

Figure 1: The running configuration of Router0

Under the interface FastEthernet0/0, the IP address 192.168.1.254/24 (subnet mask: 255.255.255.0 = /24) is configured in this interface. PC0, PC1, and PC2 need to obtain IP addresses from a remote DHCP server located on a different network. Since router does not forward broadcast packets by default, the “ip helper-address 201.1.1.2” is required to forward DHCP broadcast requests to the DHCP server. The “ip nat inside” is used to configure this interface as the private network side.

For the interface FastEthernet0/1, the IP address 201.1.1.1 with a subnet mask of /24 is configured. This is also the public IP address for the private network. The “ip nat outside” is used to configure this interface as the public network side.

“ip nat inside source list 10 interface FastEthernet 0/1 overload” is a command that enable port address translation (PAT) on interface mode. This means it allows multiple private IP address access to internet with using a single public IP address. So, this command will translate IP address in Access List 10 to the public IP address, 201.1.1.1 of the interface FastEthernet 0/1. The router will use port numbers to distinguish between the connections from devices.

“ip route 0.0.0.0 0.0.0.0 201.1.1.2” is a command that used to configure for default route. When there is no specific route in the routing table for a destination IP, the packets will be forwarded to 201.1.1.2.

The command “access-list 10 permit 192.168.1.0 0.0.0.255” sets up an Access Control List (ACL). The wildcard 0.0.0.255 only permits traffic from the 192.168.2.0/24 subnet. Hence, IP range from 192.168.1.0 to 192.168.1.254 will be included in the PAT process.

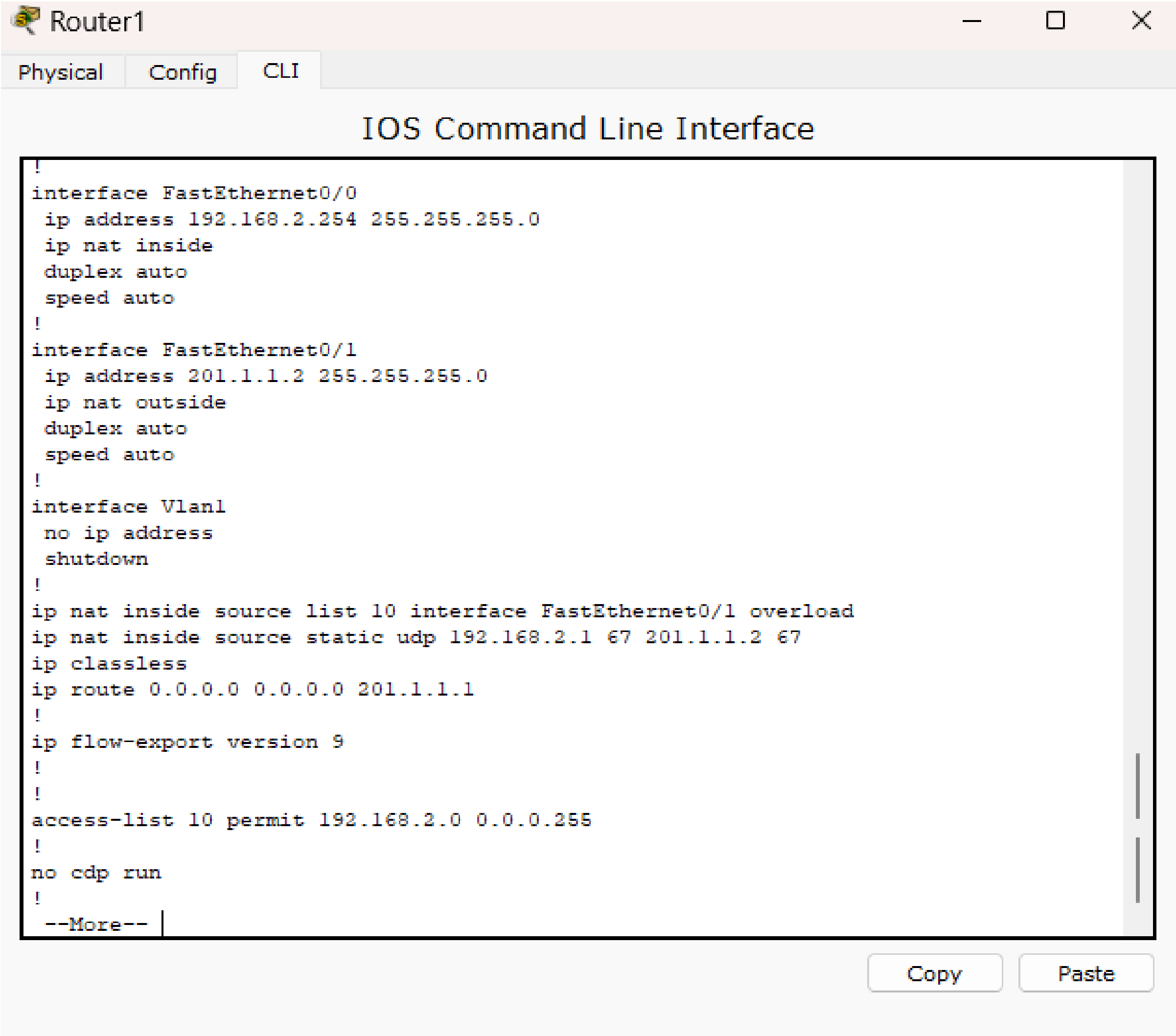


Figure 2: The running configuration of Router1

Under the interface FastEthernet 0/0, the IP address 192.168.2.254/24 (subnet mask: 255.255.255.0 = /24) is configured. The “ip nat inside” is used to configure this interface as the private network site.

For the interface FastEthernet 0/1, the IP address 201.1.1.2 with a subnet mask of /24 is configured in this interface. This is also the public IP address for the private network. The “ip nat outside” is used to configure this interface as the public network site.

“ip nat inside source list 10 interface FastEthernet 0/1 overload” is a command that enable port address translation (PAT) on interface mode. This means it allows multiple private IP address access to internet with using a single public IP address. So, this command will translate IP address in Access List 10 to the public IP address, 201.1.1.2 of the interface FastEthernet0/1. Port number will be used by the router to distinguish the connection between devices "ip route 0.0.0.0 0.0.0.0 201.1.1.1" is a command that used to configure for default route. When there is no specific route in the routing table for a destination IP, the packets will be forwarded to 201.1.1.1.

"ip nat inside source static udp 192.168.2.1 67 201.1.1.2 67" is used for port forwarding. UDP port 67 is used for exchanging DHCP message between client and server. Therefore, when packets with destination port 67 reaches this interface with the public IP address of 201.1.1.2, they will be redirected to 192.168.2.1, which the IP address of the DHCP Server 0.

The command "access-list 10 permit 192.168.2.0 0.0.0.255" sets up an Access Control List (ACL). The wildcard 0.0.0.255 only permits traffic from the 192.168.2.0/24 subnet. Hence, IP range from 192.168.2.0 to 192.168.2.254 will be included in the PAT process.

Question B

PC0:

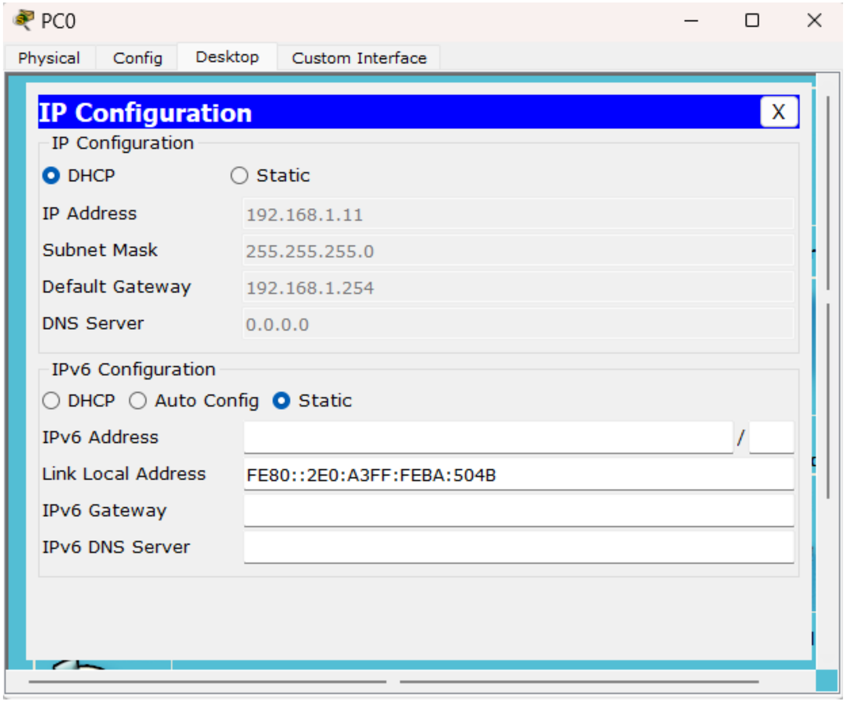


Figure 3: PC0 receives IP address via DHCP

PC1:

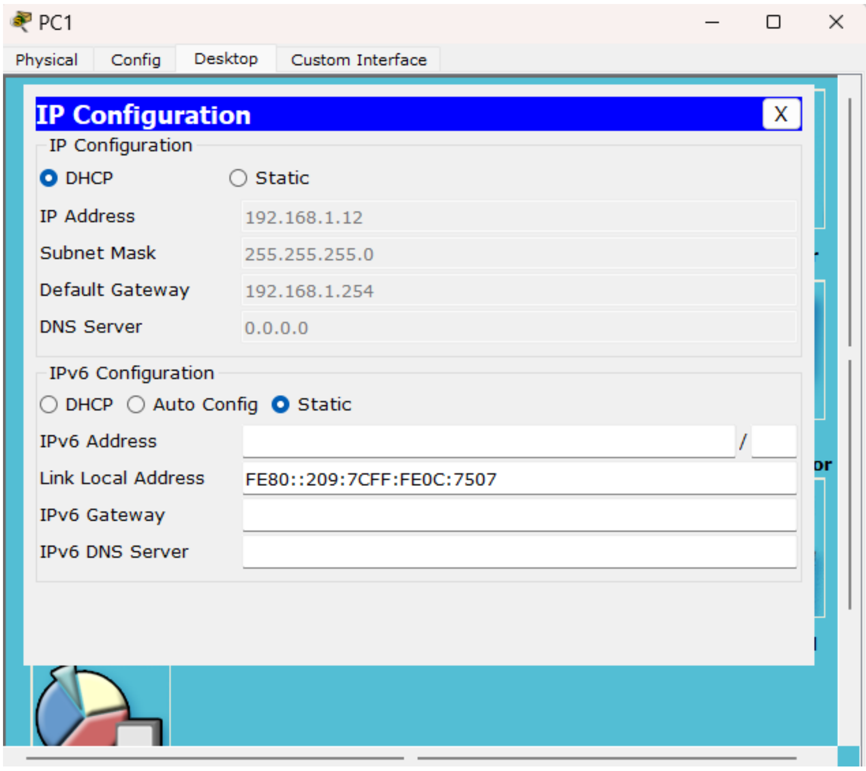


Figure 4: PC1 receives IP address vis DHCP

PC2:

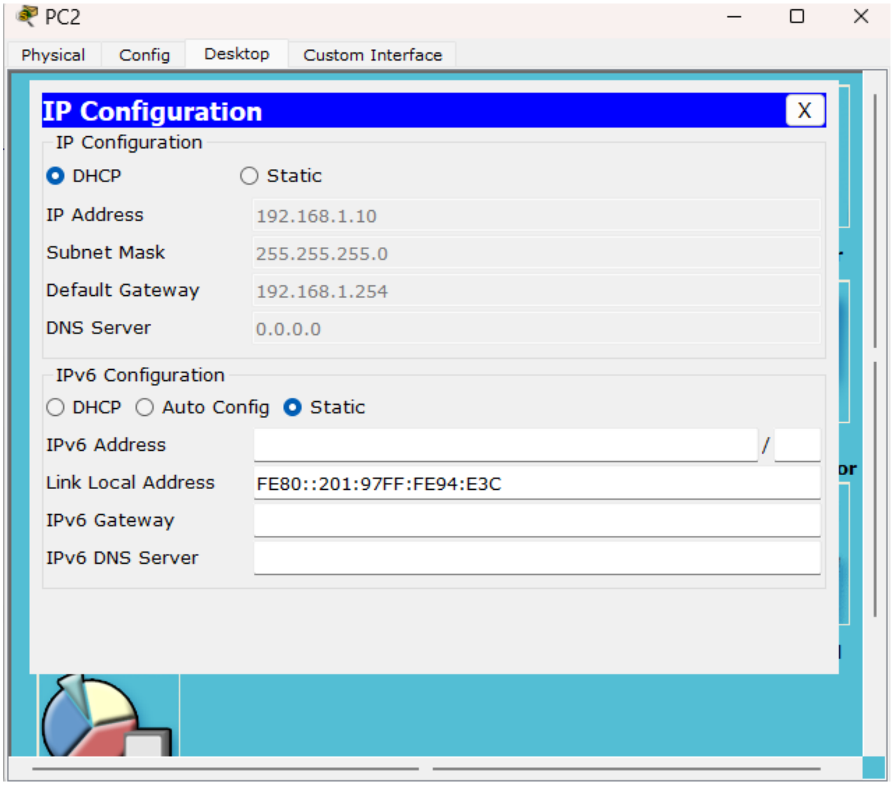


Figure 5: PC2 receives IP address via DHCP

The above 3 figures show that the PC0, PC1 and PC2 have received their IP addresses from DHCP Server0. The DHCP discover packets (broadcast packet) are sent by these three PCs. With the help of "ip helper-address" to act as DHCP relay agent, Router0 forwards the packets to another subnet, 201.1.1.2. Based on the figure 7, the DHCP discover packets are received by Router1 and the IP address is translated from 201.1.1.2 to the IP address of the DHCP Server0, 192.168.2.1.

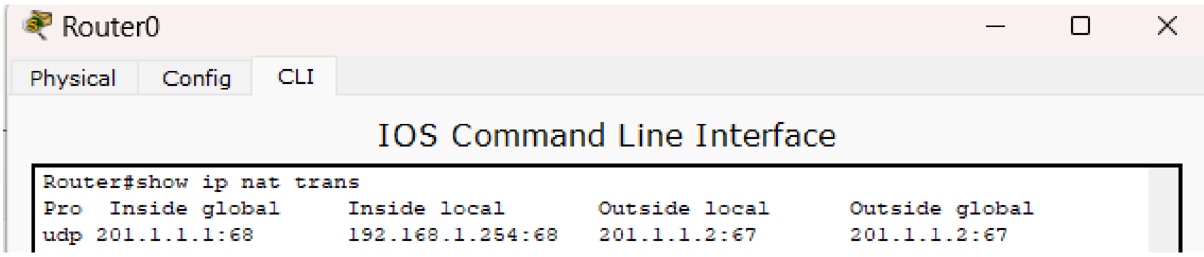


Figure 6: NAT translation table of Router0

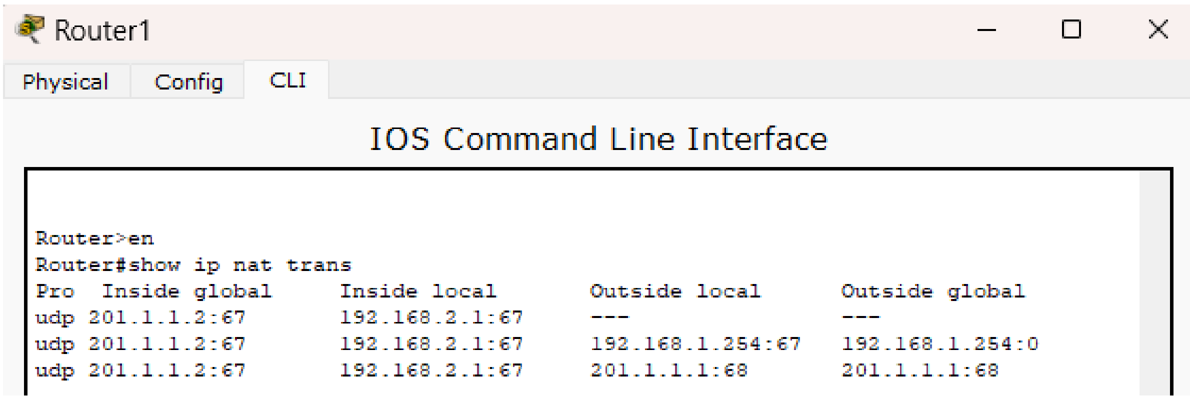


Figure 7: NAT translation table of Router1

Based on Figure 6 and Figure 7, both figures show the NAT translation tables on Router0 and Router1 respectively. When Router1 receives DHCP packets, it translates the IP address and forward the request packet to DHCP Server0 at 192.168.2.1 via port 67. After that, the DHCP offer packets are sent back to Router0 from Router1. Router0 also perform NAT translation to send the DHCP reply to the corresponding PC. The remaining DHCP packets will follow the similar paths until all PCs obtain their IP address.

Question C

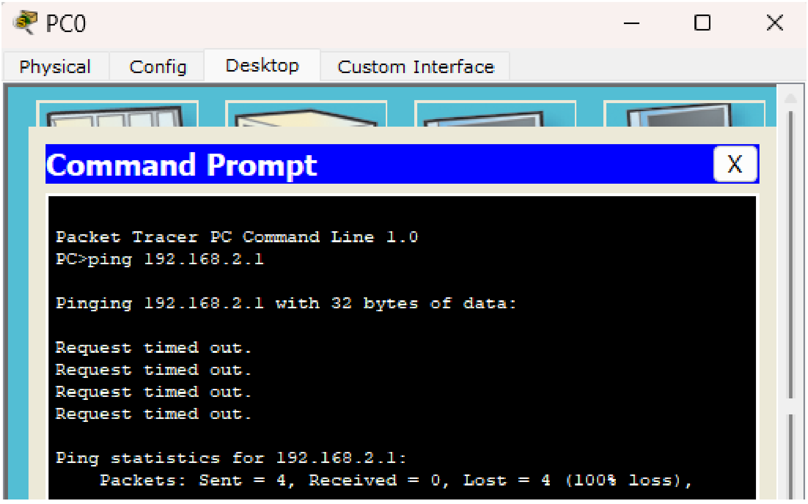


Figure 8: PC0 failed to ping Server0

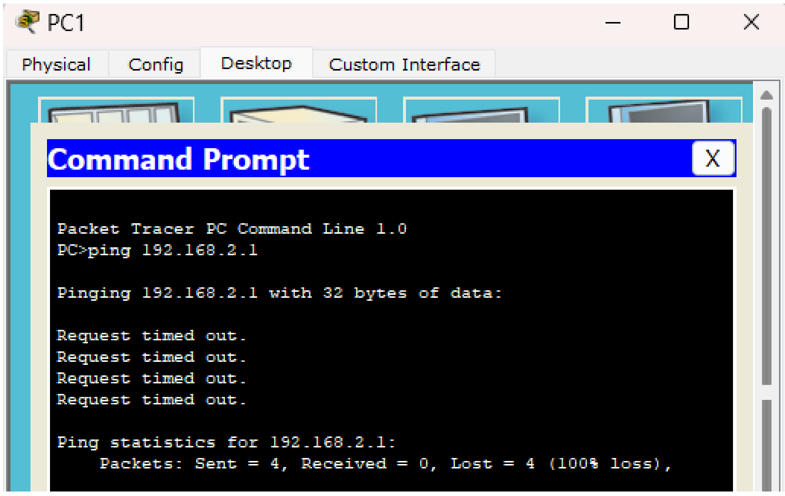


Figure 9: PC1 failed to ping Server0

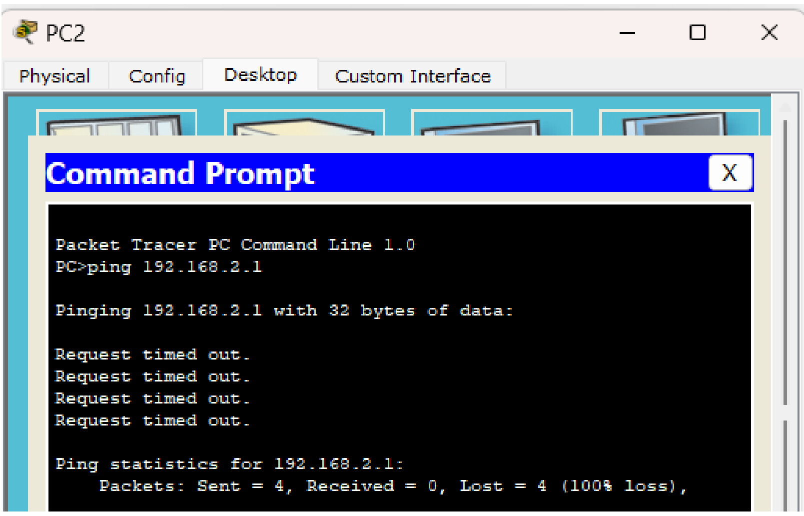


Figure 10: PC2 failed to ping Server0

No, PC0, PC1 and PC2 can’t ping directly to the DHCP Server0. This is because PCs are in one private subnet, while DHCP Server0 on another private subnet. By default, devices in inside network cannot communicate directly with devices from outside network without any proper routing or network address translation. Moreover, there is no NAT configuration at Router1 to forward ICMP (ping) packets to DHCP Server0. This causes Router1 will drop the packets, as it does not know how to handle them. Therefore, the PCs received timeout replies as their ICMP packets cannot reach the DHCP Server0.

Question 2

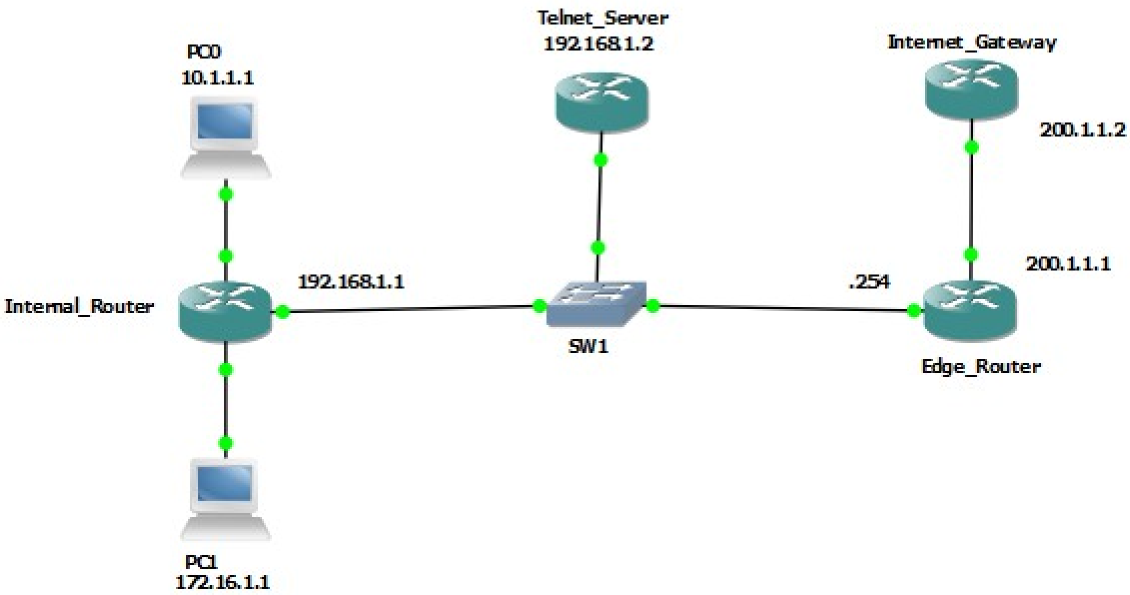


Figure 11: Network Topology

Question A

```
interface FastEthernet0/0
 ip address 200.1.1.1 255.255.255.0
 ip nat outside
 ip virtual-reassembly
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 192.168.1.254 255.255.255.0
 ip nat inside
 ip virtual-reassembly
 duplex auto
 speed auto
!
ip forward-protocol nd
!
!
ip http server
no ip http secure-server
ip nat inside source list 10 interface FastEthernet0/0 overload
ip nat inside source static tcp 192.168.1.2 23 200.1.1.1 23 extendable
!
access-list 10 permit 192.168.1.0 0.0.0.255
```

Figure 12: The running configuration of edge router

The IP address is configured as 200.1.1.1/24 (subnet mask: 255.255.255.0 = /24) on the FastEthernet0/0 interface. It is also configured as public network site with the command “ip nat outside”.

The IP address 192.168.1.254 with a subnet mask of /24 is assigned as the IP address for the FastEthernet0/1 interface. It is also configured as private network site with the command “ip nat inside”.

An Access Control List (ACL) 10 is created with the command of “access-list 10 permit 192.168.1.0 0.0.0.255”, where the wildcard mask 0.0.0.255 only permits traffic from the 192.168.1.0/24 subnet. This includes IP range from 192.168.1.1 to 192.168.1.254. “ip nat inside source list 10 interface FastEthernet0/0 overload” is the command to enable Port Address Translation (PAT) in Edge Router. It allows multiple private IP addresses share a public address. Therefore, the permitted IP addresses will share the same public address assigned to interface FastEthernet0/0, which is 200.1.1.1 .

“ip nat inside source static tcp 192.168.1.2 23 200.1.1.1 23 extendable” is a command that used to configure port forwarding on a router. Any TCP packets with destination port 23 (Telnet) reach the interfaces with public IP address 201.1.1.1 will be forwarded to the device, 192.168.1.2. The extended keyword is used to map a private IP address to multiple public IP addresses.

Question B

```
interface FastEthernet0/0
 ip address 10.1.1.254 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 172.16.1.254 255.255.255.0
 ip nat inside
 ip virtual-reassembly
 duplex auto
 speed auto
!
interface FastEthernet1/0
 ip address 192.168.1.1 255.255.255.0
 ip nat outside
 ip virtual-reassembly
 duplex auto
 speed auto
!
ip forward-protocol nd
ip route 200.1.1.0 255.255.255.0 192.168.1.254
!
!
ip http server
no ip http secure-server
ip nat inside source list 10 interface FastEthernet1/0 overload
!
access-list 10 permit 172.16.1.1
```

Figure 13: The running configuration of Internal Router

Under FastEthernet0/0, IP address 10.1.1.254/24 (subnet mask: 255.255.255.0 = /24) is assigned to this interface.

For FastEthernet0/1, IP address 172.16.1.254 with a subnet mask of /24 is assigned to this interface. It has been set up as a private network site with “ip nat inside”.

The IP address 192.168.1.1 with a subnet mask of /24 is configured for the interface FastEthernet1/0. This interface is set as public network site with the command “ip nat outside”.

“ip route 200.1.1.0 255.255.255.0 192.168.1.254” is a command that used to create static route on router. Therefore, when router receive packets with its’ destination as the 200.1.1.0/24 network, it will forward these packets to the next-hop IP address 192.168.1.254

“access-list 10 permit 172.16.1.1” is configured to create an Access Control List (ACL) and it only allows traffic from IP address 172.16.1.1 . “ip nat inside source list 10 interface FastEthernet0/0 overload” is the command to enable Port Address Translation (PAT) in Internal Router. It allows multiple private IP addresses share a public address. Therefore, the permitted

IP addresses will share the same public address assigned to interface FastEthernet1/0, which is 192.168.1.1.

Question C

Question I

PC1 ping Internet Gateway

933	4925.27700	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=1/256, ttl=64
934	4925.35200	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=1/256, ttl=253
935	4926.27900	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=2/512, ttl=64
936	4926.35600	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=2/512, ttl=253
937	4927.28300	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=3/768, ttl=64
938	4927.35100	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=3/768, ttl=253
939	4928.28900	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=4/1024, ttl=64
940	4928.36100	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=4/1024, ttl=253

Figure 14: ICMP packets flow between PC1 to Internal Router

Based on the Figure 13, it shows that PC1 (172.16.1.1) can ping the Internal Gateway (200.1.1.2). This is because there are ICMP echo requests which sent by PC1 to the Internal Gateway, while the Internal Gateway replies an ICMP echo reply for each request from PC1. The sequence number in the request packet and reply packet are the same. This means that there are no packets lost during the transmission.

3	4.016000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=1/256, ttl=63
4	4.062000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=1/256, ttl=254
5	5.021000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=2/512, ttl=63
6	5.068000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=2/512, ttl=254
7	6.026000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=3/768, ttl=63
8	6.073000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=3/768, ttl=254
9	7.036000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=4/1024, ttl=63
10	7.083000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=4/1024, ttl=254

Figure 15: ICMP packets flow between Internal Router to Edge Router

Looking at the Figure 14, the private address of PC1 is translated from 172.16.1.1 to 192.168.1.1. This is also the public IP address for the outside network on Internal Router. Since the sequence number of each reply matches the corresponding request, no packets are lost.

After that, the public IP address of PC1 is translated from 192.168.1.1 to 200.1.1.1. It can send

1340	3095.82000	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=1/256, ttl=62
1341	3095.83600	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=1/256, ttl=255
1342	3096.81700	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=2/512, ttl=62
1343	3096.83300	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=2/512, ttl=255
1344	3097.82600	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=3/768, ttl=62
1345	3097.84200	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=3/768, ttl=255
1346	3098.83100	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=4/1024, ttl=62
1347	3098.84700	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=4/1024, ttl=255

Figure 16: ICMP packets flow between Edge Router to Internet Gateway

ICMP echo request to the Internet Gateway. At the same time, Internet Gateway replies with ICMP echo reply to each request. As the sequence number for every pair of reply and request packet is the same, so there are no packets are lost.

PC1 ping PC0

959	5003.53800	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) request id=0x8b09, seq=1/256, ttl=64
960	5003.56300	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8b09, seq=1/256, ttl=63
961	5004.54200	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) request id=0x8b09, seq=2/512, ttl=64
962	5004.57400	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8b09, seq=2/512, ttl=63
963	5005.54800	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) request id=0x8b09, seq=3/768, ttl=64
964	5005.56600	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8b09, seq=3/768, ttl=63
965	5006.55400	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) request id=0x8b09, seq=4/1024, ttl=64
966	5006.57400	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8b09, seq=4/1024, ttl=63

Figure 17: PC1 successfully ping PC0

PC1 and PC0 are in the different private networks that are connected to a common router, Internal Router. Hence, there is no need for Network Address Translation (NAT). Based on the figure 16, PC1 (172.16.1.1) can successfully ping PC0 (10.1.1.1) with any packets lost.

PC1 ping Telnet Server

970	5033.52900	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0x8d09, seq=1/256, ttl=64
971	5033.56800	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8d09, seq=1/256, ttl=254
972	5034.53100	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0x8d09, seq=2/512, ttl=64
973	5034.57600	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8d09, seq=2/512, ttl=254
974	5035.53600	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0x8d09, seq=3/768, ttl=64
975	5035.57800	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8d09, seq=3/768, ttl=254
976	5036.54300	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0x8d09, seq=4/1024, ttl=64
977	5036.58600	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply id=0x8d09, seq=4/1024, ttl=254

Figure 18: ICMP packets flow between PC1 to Edge Router

From the Figure 17, PC1 is able to send ICMP echo requests to the Telnet Server, and the Telnet Server can reply each request with an ICMP echo reply.

114	311.314000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xa109, seq=1/256, ttl=63
115	311.330000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0xa109, seq=1/256, ttl=255
116	311.471000	c4:02:51:94:00:10	c4:02:51:94:00:10	LOOP	60	Reply
117	312.309000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xa109, seq=2/512, ttl=63
118	312.325000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0xa109, seq=2/512, ttl=255
119	313.319000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xa109, seq=3/768, ttl=63
120	313.334000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0xa109, seq=3/768, ttl=255
121	314.327000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xa109, seq=4/1024, ttl=63
122	314.342000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply id=0xa109, seq=4/1024, ttl=255

Figure 19: ICMP packets flow between Edge Router to Telnet Server

However, PC1 is in the private network site, while the Telnet Server is in the public network site. Hence, Network Address Translation (NAT) is required. The private IP address of PC1, 172.16.1.1 is translated to public IP address, 192.168.1.1. As a result, PC1 can successfully ping Telnet Server without any packet loss.

Question II

PC0 unable to ping Internet Gateway

28	105.675000	10.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request id=0xab09, seq=1/256, ttl=64
29	106.690000	10.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request id=0xab09, seq=2/512, ttl=64
30	107.690000	10.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request id=0xab09, seq=3/768, ttl=64
31	108.690000	10.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request id=0xab09, seq=4/1024, ttl=64

Figure 20: PC0 failed to ping Internet Gateway

Based on Figure 19, PC0 (10.1.1.1) is unable to ping Internet Gateway (200.1.1.2). This can be observed as there is no ICMP echo reply replied from Internet Gateway after PC0 sent the ICMP echo requests to it. This issue is caused by the configuration of the Access Control List (ACL) on the edge router. It only permits traffic from the 192.168.1.0/24 network. Since PC0 IP address is 10.1.1.1, its traffic is denied by the ACL. Additionally, the internal router is configured to translate only the IP address 172.16.1.1 to the public IP address 192.168.1.1. Hence, the IP address of PC0 is not translated to 192.168.1.1. As a result, the ICMP echo requests from PC0 are blocked.

PC0 ping PC1

38	138.332000	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) request id=0xac09, seq=1/256, ttl=64
39	138.357000	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) reply id=0xac09, seq=1/256, ttl=63
40	139.338000	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) request id=0xac09, seq=2/512, ttl=64
41	139.364000	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) reply id=0xac09, seq=2/512, ttl=63
42	140.342000	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) request id=0xac09, seq=3/768, ttl=64
43	140.372000	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) reply id=0xac09, seq=3/768, ttl=63
44	141.346000	10.1.1.1	172.16.1.1	ICMP	98	Echo (ping) request id=0xac09, seq=4/1024, ttl=64
45	141.367000	172.16.1.1	10.1.1.1	ICMP	98	Echo (ping) reply id=0xac09, seq=4/1024, ttl=63

Figure 21: PC0 successfully ping PC1

PC0 and PC1 are in the different private networks that are connected to a common router. Hence, there is no need for Network Address Translation (NAT). Based on the figure 21, PC0 (172.16.1.1) can successfully ping PC1 (10.1.1.1) with any packets lost.

PC0 ping Telnet Server

48	158.245000	10.1.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xad09, seq=1/256, ttl=64
49	158.294000	192.168.1.2	10.1.1.1	ICMP	98	Echo (ping) reply id=0xad09, seq=1/256, ttl=254
50	159.252000	10.1.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xad09, seq=2/512, ttl=64
51	159.287000	192.168.1.2	10.1.1.1	ICMP	98	Echo (ping) reply id=0xad09, seq=2/512, ttl=254
52	160.258000	10.1.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xad09, seq=3/768, ttl=64
53	160.297000	192.168.1.2	10.1.1.1	ICMP	98	Echo (ping) reply id=0xad09, seq=3/768, ttl=254
54	161.263000	10.1.1.1	192.168.1.2	ICMP	98	Echo (ping) request id=0xad09, seq=4/1024, ttl=64
55	161.307000	192.168.1.2	10.1.1.1	ICMP	98	Echo (ping) reply id=0xad09, seq=4/1024, ttl=254

Figure 22: ICMP packets flow between PC0 to Internal Router

Looking at Figure 21, PC0 can send ICMP echo requests to the Telnet Server, and the Telnet Server can reply each request with an ICMP echo reply. This indicates that PC0 can successfully ping Telnet Server without any packet loss.

Question III

The NAT occurs during PC1 ping Internet Gateway

933	4925.27700	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=1/256, ttl=64
934	4925.35200	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=1/256, ttl=253
935	4926.27900	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=2/512, ttl=64
936	4926.35600	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=2/512, ttl=253
937	4927.28300	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=3/768, ttl=64
938	4927.35100	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=3/768, ttl=253
939	4928.28900	172.16.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x8909, seq=4/1024, ttl=64
940	4928.36100	200.1.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8909, seq=4/1024, ttl=253

Figure 23: ICMP packets flow between PC1 to Internal Router

3	4.016000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=1/256, ttl=63
4	4.062000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=1/256, ttl=254
5	5.021000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=2/512, ttl=63
6	5.068000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=2/512, ttl=254
7	6.026000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=3/768, ttl=63
8	6.073000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=3/768, ttl=254
9	7.036000	192.168.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0x9309, seq=4/1024, ttl=63
10	7.083000	200.1.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0x9309, seq=4/1024, ttl=254

Figure 24: ICMP packets flow between Internal Router to Edge Router

1340	3095.82000	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=1/256, ttl=62
1341	3095.83600	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=1/256, ttl=255
1342	3096.81700	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=2/512, ttl=62
1343	3096.83300	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=2/512, ttl=255
1344	3097.82600	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=3/768, ttl=62
1345	3097.84200	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=3/768, ttl=255
1346	3098.83100	200.1.1.1	200.1.1.2	ICMP	98	Echo (ping) request	id=0xa009, seq=4/1024, ttl=62
1347	3098.84700	200.1.1.2	200.1.1.1	ICMP	98	Echo (ping) reply	id=0xa009, seq=4/1024, ttl=255

Figure 25: ICMP packets flow between Edge Router to Internet Gateway

Based on the figure above, PC1's private IP address has been translated from 172.16.1.1 to 192.168.1.1, and then to 200.1.1.1. This is because PC1 is in a private network and Internet Gateway is at the public network site. Additionally, the destination IP address in the return packets is also translated from 200.1.1.1 to 192.168.1.1, and finally to 172.16.1.1. This ensures that the reply reaches PC1 correctly. The double translation occurs from the implementation of double NAT, once from the edge router and another time from the internal router.

The NAT occurs during the PC1 ping Telnet Server

970	5033.52900	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0x8d09, seq=1/256, ttl=64
971	5033.56800	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8d09, seq=1/256, ttl=254
972	5034.53100	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0x8d09, seq=2/512, ttl=64
973	5034.57600	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8d09, seq=2/512, ttl=254
974	5035.53600	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0x8d09, seq=3/768, ttl=64
975	5035.57800	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8d09, seq=3/768, ttl=254
976	5036.54300	172.16.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0x8d09, seq=4/1024, ttl=64
977	5036.58600	192.168.1.2	172.16.1.1	ICMP	98	Echo (ping) reply	id=0x8d09, seq=4/1024, ttl=254

Figure 26: ICMP packets flow between PC1 to Internal Router

114	311.314000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0xa109, seq=1/256, ttl=63
115	311.330000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0xa109, seq=1/256, ttl=255
116	311.471000	c4:02:51:94:00:10	c4:02:51:94:00:10	LOOP	60	Reply	
117	312.309000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0xa109, seq=2/512, ttl=63
118	312.325000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0xa109, seq=2/512, ttl=255
119	313.319000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0xa109, seq=3/768, ttl=63
120	313.334000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0xa109, seq=3/768, ttl=255
121	314.327000	192.168.1.1	192.168.1.2	ICMP	98	Echo (ping) request	id=0xa109, seq=4/1024, ttl=63
122	314.342000	192.168.1.2	192.168.1.1	ICMP	98	Echo (ping) reply	id=0xa109, seq=4/1024, ttl=255

Figure 27: ICMP packets flow between Internal Router to Telnet Server

From the figure above, the private IP address of PC1 has been translated from 172.16.1.1 to 192.168.1.1. After that, the return packets' destination IP address will be translated from 192.168.1.1 to 172.16.1.1.

Question D

```
Router#telnet 200.1.1.1
Trying 200.1.1.1 ... Open


User Access Verification

Password:
Router>exit

[Connection to 200.1.1.1 closed by foreign host]
Router#ping 192.168.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
Router#
```

Figure 28: Telnet session established, and ping session failed

1579	3887.26900	200.1.1.2	200.1.1.1	TCP	60	59852 > telnet [ACK] Seq=1 Ack=1 Win=4128 Len=0
1580	3887.28400	200.1.1.2	200.1.1.1	TELNET	63	Telnet Data ...
1581	3887.30000	200.1.1.2	200.1.1.1	TCP	60	[TCP Dup ACK 1580#1] 59852 > telnet [ACK] Seq=10 Ack=1 Win=4128 Len=0
1582	3887.33200	200.1.1.1	200.1.1.2	TELNET	66	Telnet Data ...
1583	3887.34700	200.1.1.2	200.1.1.1	TELNET	60	Telnet Data ...
1584	3887.34700	200.1.1.1	200.1.1.2	TELNET	96	Telnet Data ...
1585	3887.36300	200.1.1.2	200.1.1.1	TELNET	60	Telnet Data ...
1586	3887.36300	200.1.1.1	200.1.1.2	TELNET	57	Telnet Data ...
1587	3887.37900	200.1.1.2	200.1.1.1	TELNET	63	Telnet Data ...
1588	3887.37900	200.1.1.1	200.1.1.2	TELNET	60	Telnet Data ...
1589	3887.42600	200.1.1.1	200.1.1.2	TELNET	57	Telnet Data ...
1590	3887.55200	200.1.1.2	200.1.1.1	TCP	60	59852 > telnet [ACK] Seq=25 Ack=67 Win=4062 Len=0
1591	3887.64700	200.1.1.1	200.1.1.2	TCP	54	telnet > 59852 [ACK] Seq=67 Ack=25 Win=4104 Len=0
1592	3888.49800	c4:03:51:94:00:00	CDP/VTP/DTP/PAgP/UD	CDP	363	Device ID: Router Port ID: FastEthernet0/0
1593	3888.68700	200.1.1.2	200.1.1.1	TELNET	60	Telnet Data ...
1594	3888.76600	200.1.1.2	200.1.1.1	TELNET	60	Telnet Data ...
1595	3888.84500	200.1.1.2	200.1.1.1	TELNET	60	Telnet Data ...

Figure 29: Continuous Telnet packet exchange

From Figure 29, it shows that the Internet Gateway can connect to the Telnet Server using telnet with the command “telnet 200.1.1.1”. The telnet session is successfully established. This is because port forwarding is configured in Edge Router, which allows packets with destination port 23 will be redirected to the Telnet Server, 192.168.1.2 after reaching the Edge Router’s interface, 200.1.1.1. Therefore, Edge Router can forward the Telnet packets from outside network to the Telnet Server that located in the private network site. However, Internet Gateway cannot ping the Telnet Server directly. It shows the success rate is 0 percent. This is because the Edge Router is configured only to forward TCP traffic on port 23 (Telnet) and does not forward ICMP packets. As a result, Telnet connection is allowed, but the ICMP packets will be dropped off.

Looking at the Wireshark capture, the Telnet session is successfully established. Internet Gateway (200.1.1.2) and Telnet Server with public IP address (200.1.1.1) are communicating with each other. This can be showed by continuous packets exchange between these two devices.